FOUR EQUIVALENT FORMULATIONS OF SEPARATEDNESS FOR A MAP OF SPECTRAL SPACES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of the diagonal of a map of spectral spaces through the topology of the fibre product. Used inside the counted proof (source lines 432-440). Turns the diagonal being a closed immersion into a condition on the diagonal as a subspace.

Source reference. Original definition definition-separated: sources/stacks-topology/pinned-original-source.tex, source lines 413–413. Navigation pointer only: the counted text refers to the Stacks item definition-separated, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : X \to Y$ be continuous map of topological spaces. The following are equivalent:*

1. *f is separated,*
2. *$\Delta(X) \subset X \times_Y X$ is a closed subset,*
3. *given distinct points $x, x' \in X$ mapping to the same point of Y, there exist disjoint open neighbourhoods of x and x'.*

Proof. If f is separated, by Definition definition-separated, Δ is a closed map. The fact that X is closed in X gives us that $\Delta(X)$ is closed in $X \times_Y X$. Thus (1) implies (2).

Assume $\Delta(X) \subset X \times_Y X$ is a closed subset and let U denote the complementary open. This means we have an open set $W \subset X \times X$ such that $W \cap (X \times_Y X) = U$. However, by definition of the product topology, if $(x, x') \in W \cap (X \times_Y X)$, we have V and V' open sets of X such that $x \in V$, $x' \in V'$ and $V \times V' \subset W$. If

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we had $V \cap V' \neq \emptyset$, we would have $z \in V \cap V'$. However, $(z, z) \in X \times_Y X$, so $(z, z) \in (V \times V') \cap (X \times_Y X) \subset U$, which is absurd. Therefore $V \cap V' = \emptyset$, and we have two disjoint open neighborhoods for x and x'. It proves that (2) implies (3).

Finally, we suppose that given distinct points $x, x' \in X$ mapping to the same point of Y, there exist disjoint open neighbourhoods of x and x'. Let F be a closed set of X. We will show that $\Delta(F)$ is a closed subset of $X \times_Y X$. Let $(x, x') \in X \times_Y X$ be a point not contained in $\Delta(F)$. Then either $x \neq x'$ or $x \notin F$. In the first case, we choose disjoint open neighbourhoods $V, V' \subset X$ of x, x' and we see that $(V \times V') \cap X \times_Y X$ is an open neighbourhood of (x, x') not meeting $\Delta(F)$. In the second case, we see that $((X \setminus F) \times X) \cap X \times_Y X$ is an open neighbourhood of (x, x') not meeting $\Delta(F)$. We have shown that (3) implies (1). $\square$